

Codeforces Practice Sets

Recent contest problems (mostly 2024-2026), Specialist+ band. Grouped into sets. Topics and ratings are hidden until the answer key. Figure out the pattern yourself.

Calibrated against Codeforces handle Oju (Specialist, max 1419): excludes problems already AC'd on that account.

How to open a problem

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2230D -> https://codeforces.com/problemset/problem/2230/D
2149E -> https://codeforces.com/problemset/problem/2149/E
Format: /problemset/problem/{contestId}/{index}
```

Rules

- No AI until AFTER you submit (or timebox ends).
- Timer: 40-70 min per problem. These are not Div2 A toys.
- Before code: guess the pattern in one line. Write what you track.
- Stuck 25 min: write a correct slow solution, then optimize.
- Editorial only after a real attempt. Re-solve fails cold in 3 days.
- Finish one full set before the next. Do NOT open the answer key early.

One problem flow

1. Guess pattern (2 min)
2. What state / data structure? (3 min)
3. Dry-run samples + edge cases (5 min)
4. Code + stress tiny cases if needed
5. Submit. Editorial only for gaps
6. Log: guessed pattern | key topic | bug

Set 1

- 2239B Decidophobia
- 2234F Vessels, Heights and Two Versions (Hard Version)
- 2233D Goods on the Shelf
- 2232C2 Seating Arrangement (Hard Version)
- 2226C Mental Monumental (Easy Version)
- 2223B Zhily and Barknights
- 2196B Another Problem about Beautiful Pairs
- 2185E The Robotic Rush

Set 2

- 2229D Me When Median Problem
- 2228D Sanae, Cross and Color
- 2203C Test Generator
- 2183C War Strategy
- 2182E New Year's Gifts
- 2169D1 Removal of a Sequence (Easy Version)
- 2167E khba Loves to Sleep!
- 2157D Billion Players Game

Set 3

- 2242D Two Digit Strings
 - 2241E Fair and Square
 - 2234D XOR, Expression and Two Binary Numbers
 - 2230D Good Schedule
 - 2228C1 Cirno and Number (Easy Version)
 - 2227F It Just Keeps Going Sideways
 - 2226D Reserved Reversals
 - 2222C Median Partition
-

Set 4

- 2211C2 Equal Multisets (Hard Version)
 - 2208D1 Tree Orientation (Easy Version)
 - 2184E Exquisite Array
 - 2183D1 Tree Coloring (Easy Version)
 - 2172F Cluster Computing System
 - 2154D Catshock
 - 2127D Root was Built by Love, Broken by Destiny
 - 2117F Wildflower
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Set 5

- 2218F The 67th Tree Problem
 - 2205D Simons and Beating Peaks
 - 2195E Idiot First Search
 - 2192D Cost of Tree
 - 2184F Cherry Tree
 - 2171F Rae Taylor and Trees (hard version)
 - 2167F Tree, TREE!!!
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Set 6

- 2238D Storming Arasaka
 - 2237D Fullmetal Bitchemist
 - 2217D Flip the Bit (Hard Version)
 - 2209D Ghostfires
 - 2207B One Night At Freddy's
 - 2205C Simons and Posting Blogs
 - 2203D Divisibility Game
 - 2200E Divisive Battle
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Set 7

- 2241F A Bit Odd
 - 2222D Permutation Construction
 - 2200F Mooclear Reactor 2
 - 2185F BattleCows
 - 2149E Hidden Knowledge of the Ancients
 - 2138B Antiamuny Wants to Learn Swap
 - 2137F Prefix Maximum Invariance
 - 2132E Arithmetics Competition
-

Set 8

- 2190B2 Sub-RBS (Hard Version)
 - 2164D Copy String
 - 2121E Sponsor of Your Problems
 - 2111E Changing the String
 - 2110D Fewer Batteries
 - 2104E Unpleasant Strings
 - 2092D Mishkin Energizer
 - 2086D Even String
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Set 9

- 2225D Exceptional Segments
 - 2211D AND-array
 - 2189C2 XOR-convenience (Hard Version)
 - 2146D1 Max Sum OR (Easy Version)
 - 2093F Hackers and Neural Networks
 - 2072F Goodbye, Banker Life
 - 2061D Kevin and Numbers
 - 2057C Trip to the Olympiad
-

Set 10

- 2236F1 Elections in Saransk (easy version)
 - 2204E Sum of Digits (and Again)
 - 2179E Blackslax and Girls
 - 2178D Xmas or Hysteria
 - 2163B Siga ta Kymata
 - 2152D Division Versus Addition
 - 2150B Grid Counting
 - 2149F Nezuko in the Clearing
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Set 11

- 2201B Recollect Numbers
 - 2194D Table Cut
 - 2162E Beautiful Palindromes
 - 2153C Symmetrical Polygons
 - 2123F Minimize Fixed Points
 - 2122C Manhattan Pairs
 - 2113C Smilo and Minecraft
 - 2094F Trulimero Trulicina
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Set 12

- 2181F Fragmented Nim
 - 2161B Make Connected
 - 2155C The Ancient Wizards' Capes
 - 2148F Gravity Falls
 - 2147D Game on Array
 - 2137E Mexification
 - 2121F Yamakasi
 - 2117E Lost Soul
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ANSWER KEY - open only after practice

Topic + rating per set. Use to check pattern recognition after you solve, not to cheat the technique beforehand.

Band: roughly 1500-1900 CF. Source: recent problemset, unsolved by Oju.

Set 1: Two Pointers / Sliding Window

- 2239B (1500) Decidophobia
tags: greedy, math, sortings, two pointers
 - 2234F (1900) Vessels, Heights and Two Versions (Hard Version)
tags: data structures, dfs and similar, dsu, greedy
 - 2233D (1900) Goods on the Shelf
tags: brute force, data structures, implementation, two pointers
 - 2232C2 (1500) Seating Arrangement (Hard Version)
tags: binary search, greedy, math, two pointers
 - 2226C (1500) Mental Monumental (Easy Version)
tags: binary search, data structures, greedy, math
 - 2223B (1600) Zhily and Barknights
tags: binary search, data structures, dp, math
 - 2196B (1600) Another Problem about Beautiful Pairs
tags: brute force, math, two pointers
 - 2185E (1500) The Robotic Rush
tags: binary search, greedy, implementation, two pointers
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Set 2: Binary Search

- 2229D (1700) Me When Median Problem
tags: binary search, greedy
 - 2228D (1900) Sanae, Cross and Color
tags: binary search, data structures, implementation
 - 2203C (1500) Test Generator
tags: binary search, bitmasks, greedy, math
 - 2183C (1500) War Strategy
tags: binary search, greedy, math, two pointers
 - 2182E (1800) New Year's Gifts
tags: binary search, data structures, greedy, sortings
 - 2169D1 (1500) Removal of a Sequence (Easy Version)
tags: binary search, implementation, math, number theory
 - 2167E (1600) khba Loves to Sleep!
tags: binary search, data structures, geometry, greedy
 - 2157D (1600) Billion Players Game
tags: binary search, greedy, math, sortings
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Set 3: Dynamic Programming

- 2242D (1600) Two Digit Strings
tags: dp, strings
- 2241E (1600) Fair and Square
tags: combinatorics, dp, graphs, math
- 2234D (1500) XOR, Expression and Two Binary Numbers
tags: bitmasks, divide and conquer, dp, math
- 2230D (1700) Good Schedule
tags: dp, greedy
- 2228C1 (1500) Cirno and Number (Easy Version)
tags: binary search, brute force, dp, greedy

- 2227F (1700) It Just Keeps Going Sideways
tags: binary search, data structures, dp, greedy
 - 2226D (1900) Reserved Reversals
tags: constructive algorithms, dp, greedy, math
 - 2222C (1500) Median Partition
tags: dp, math
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Set 4: Graphs / DFS / DSU

- 2211C2 (1800) Equal Multisets (Hard Version)
tags: constructive algorithms, dsu, greedy
 - 2208D1 (1800) Tree Orientation (Easy Version)
tags: constructive algorithms, dfs and similar, dsu, graphs
 - 2184E (1800) Exquisite Array
tags: combinatorics, data structures, dsu, sortings
 - 2183D1 (1500) Tree Coloring (Easy Version)
tags: constructive algorithms, dfs and similar, greedy, trees
 - 2172F (1600) Cluster Computing System
tags: graphs, greedy, math
 - 2154D (1900) Catshock
tags: constructive algorithms, dfs and similar, graphs, trees
 - 2127D (1800) Root was Built by Love, Broken by Destiny
tags: combinatorics, dfs and similar, graphs, trees
 - 2117F (1800) Wildflower
tags: combinatorics, dfs and similar, trees
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Set 5: Trees

- 2218F (1500) The 67th Tree Problem
tags: constructive algorithms, implementation, trees
 - 2205D (1700) Simons and Beating Peaks
tags: data structures, divide and conquer, dp, greedy
 - 2195E (1500) Idiot First Search
tags: dfs and similar, dp, trees
 - 2192D (1800) Cost of Tree
tags: dp, greedy, trees
 - 2184F (1900) Cherry Tree
tags: dfs and similar, dp, graphs, trees
 - 2171F (1600) Rae Taylor and Trees (hard version)
tags: binary search, constructive algorithms, data structures, dp
 - 2167F (1600) Tree, TREE!!!
tags: dfs and similar, dp, math, trees
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Set 6: Greedy + Sortings

- 2238D (1500) Storming Arasaka
tags: greedy, math, number theory
- 2237D (1500) Fullmetal Bitchemist
tags: constructive algorithms, greedy, math
- 2217D (1900) Flip the Bit (Hard Version)
tags: greedy, implementation, math
- 2209D (1800) Ghostfires
tags: constructive algorithms, greedy
- 2207B (1600) One Night At Freddy's
tags: games, greedy, sortings

- 2205C (1500) Simons and Posting Blogs
tags: greedy, sortings
 - 2203D (1700) Divisibility Game
tags: brute force, games, greedy, number theory
 - 2200E (1500) Divisive Battle
tags: games, greedy, math, number theory
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Set 7: Data Structures

- 2241F (1700) A Bit Odd
tags: data structures, games, greedy, math
 - 2222D (1800) Permutation Construction
tags: constructive algorithms, data structures, sortings
 - 2200F (1900) Mooclear Reactor 2
tags: brute force, data structures, greedy, implementation
 - 2185F (1700) BattleCows
tags: data structures, divide and conquer, implementation
 - 2149E (1500) Hidden Knowledge of the Ancients
tags: data structures, two pointers
 - 2138B (1900) Antiamuny Wants to Learn Swap
tags: data structures, greedy, two pointers
 - 2137F (1900) Prefix Maximum Invariance
tags: binary search, combinatorics, data structures, sortings
 - 2132E (1700) Arithmetics Competition
tags: binary search, data structures, greedy, sortings
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Set 8: Strings / Hashing

- 2190B2 (1900) Sub-RBS (Hard Version)
tags: dp, games, implementation, strings
 - 2164D (1800) Copy String
tags: greedy, implementation, strings, two pointers
 - 2121E (1500) Sponsor of Your Problems
tags: dp, greedy, implementation, strings
 - 2111E (1900) Changing the String
tags: binary search, data structures, greedy, implementation
 - 2110D (1700) Fewer Batteries
tags: binary search, dfs and similar, dp, graphs
 - 2104E (1700) Unpleasant Strings
tags: binary search, dp, greedy, strings
 - 2092D (1800) Mishkin Energizer
tags: brute force, constructive algorithms, greedy, implementation
 - 2086D (1700) Even String
tags: brute force, combinatorics, dp, math
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Set 9: Bitmasks

- 2225D (1500) Exceptional Segments
tags: bitmasks, brute force, math
- 2211D (1900) AND-array
tags: bitmasks, combinatorics, math
- 2189C2 (1800) XOR-convenience (Hard Version)
tags: bitmasks, constructive algorithms, math
- 2146D1 (1500) Max Sum OR (Easy Version)
tags: bitmasks, constructive algorithms, divide and conquer, greedy

- 2093F (1800) Hackers and Neural Networks
tags: bitmasks, brute force, greedy
 - 2072F (1700) Goodbye, Banker Life
tags: 2-sat, bitmasks, combinatorics, constructive algorithms
 - 2061D (1600) Kevin and Numbers
tags: bitmasks, data structures
 - 2057C (1500) Trip to the Olympiad
tags: bitmasks, constructive algorithms, greedy, math
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Set 10: Math / Number Theory / Combinatorics

- 2236F1 (1700) Elections in Saransk (easy version)
tags: number theory
 - 2204E (1800) Sum of Digits (and Again)
tags: brute force, constructive algorithms, math
 - 2179E (1800) Blackalex and Girls
tags: constructive algorithms, geometry, math
 - 2178D (1700) Xmas or Hysteria
tags: constructive algorithms, greedy, implementation, math
 - 2163B (1700) Siga ta Kymata
tags: constructive algorithms, greedy, implementation, math
 - 2152D (1700) Division Versus Addition
tags: games, greedy, math
 - 2150B (1700) Grid Counting
tags: combinatorics, implementation, math
 - 2149F (1900) Nezuko in the Clearing
tags: binary search, math, ternary search
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Set 11: Constructive Algorithms

- 2201B (1700) Recollect Numbers
tags: constructive algorithms
 - 2194D (1600) Table Cut
tags: constructive algorithms, greedy, implementation
 - 2162E (1600) Beautiful Palindromes
tags: constructive algorithms, greedy, schedules
 - 2153C (1500) Symmetrical Polygons
tags: constructive algorithms, geometry, greedy, implementation
 - 2123F (1700) Minimize Fixed Points
tags: constructive algorithms, number theory
 - 2122C (1700) Manhattan Pairs
tags: constructive algorithms, geometry, greedy, math
 - 2113C (1700) Smilo and Minecraft
tags: brute force, constructive algorithms, greedy
 - 2094F (1600) Trulimero Trulicina
tags: constructive algorithms
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Set 12: Mixed Hard (impl / games / brute)

- 2181F (1600) Fragmented Nim
tags: games
- 2161B (1700) Make Connected
tags: brute force, implementation
- 2155C (1500) The Ancient Wizards' Capes
tags: brute force, greedy, implementation

- 2148F (1800) Gravity Falls
tags: greedy, implementation, sortings
 - 2147D (1700) Game on Array
tags: games, greedy
 - 2137E (1500) Mexification
tags: brute force, implementation, math
 - 2121F (1800) Yamakasi
tags: binary search, brute force, data structures, greedy
 - 2117E (1600) Lost Soul
tags: brute force, greedy
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If your guessed pattern matched most of the time, push Set difficulty upward (virtual Div2 C/D). If not, redo that set cold.